

$$= 24.02 = 25 \text{ N}$$

|    |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|----|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11 | B | <p>Since the total energy <math>E_T = -\frac{GMm}{2R}</math> of the satellite decreases due to work done against drag forces, its orbital radius <math>R</math> decreases.</p> <p>As <math>R</math> decreases, kinetic energy <math>E_k = \frac{GMm}{2R}</math> increases, hence orbital speed increases.</p> <p>Since <math>\frac{GMm}{R^2} = m\omega^2 R = m\left(\frac{2\pi}{T}\right)^2 R \Rightarrow T^2 \propto R^3</math></p> <p>The orbital period <math>T</math> decreases as <math>R</math> decreases.</p> |
| 12 | A | Gravitational force provides the centripetal force                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |